

ORIGINAL CONTRIBUTION

Acute Kidney Injury After Intensive Blood Pressure Lowering Following Successful Endovascular Thrombectomy

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BACKGROUND: The effect of postthrombectomy blood pressure (BP) management on the development of acute kidney injury (AKI) in patients with acute ischemic stroke remains largely unexplored.

METHODS: This secondary analysis of the OPTIMAL-BP trial (Outcome in Patients Treated With Intra-Arterial Thrombectomy-Optimal Blood Pressure Control) included patients with acute ischemic stroke due to large-vessel occlusion who achieved successful endovascular thrombectomy and had a systolic BP ≥ 140 mmHg. Patients were randomized to intensive (target systolic BP < 140 mmHg) or conventional (target systolic BP 140–180 mmHg) BP management for 24 hours. The outcomes were AKI within 7 days and within 2 days, defined according to the Kidney Disease: Improving Global Outcomes criteria. In addition, we examined the associations between AKI and functional independence at 3 months, defined as a modified Rankin Scale score of 0 to 2. Multivariable logistic regression analyses were performed with adjustment for age, sex, time from stroke onset to enrollment, baseline National Institutes of Health Stroke Scale score, and baseline estimated glomerular filtration rate.

RESULTS: Of 306 patients, 19 were excluded, and 287 patients were included in this analysis (mean age, 73.2 years; 117 [40.8%] women). AKI within 7 days occurred more frequently in the intensive management group than in the conventional group (20/147 [13.6%] versus 9/140 [6.4%]; adjusted odds ratio, 2.54 [95% CI, 1.10–6.35]). Most AKI events were stage 1 (20/29 [69.0%]). Early AKI within 2 days was also more common with intensive BP management. Patients with AKI had significantly lower rates of functional independence (4/29 [13.8%] versus 126/257 [49.0%]; adjusted odds ratio, 0.19 [95% CI, 0.05–0.55]) and higher stroke-related mortality at 3 months (11/29 [37.9%] versus 8/257 [3.1%]; adjusted odds ratio, 13.8 [95% CI, 4.14–49.64]). In a sensitivity analysis with equal creatinine ascertainment, the association with 48-hour AKI did not reach statistical significance (7/76 [9.2%] versus 2/66 [3.0%]; adjusted odds ratio, 4.20 [95% CI, 0.83–32.6]), although the absolute risk difference remained directionally consistent.

CONCLUSIONS: Intensive BP lowering after successful endovascular thrombectomy was associated with a higher risk of AKI, even when kidney injury was predominantly mild. In addition, AKI was associated with worse neurological outcomes. These findings suggest that AKI is an important marker of systemic hemodynamic vulnerability after aggressive postendovascular thrombectomy BP lowering.

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GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: acute kidney injury ■ blood pressure ■ humans ■ kidney ■ odds ratio

Nonstandard Abbreviations and Acronyms

AKI	acute kidney injury
AOR	adjusted odds ratio
BP	blood pressure
EVT	endovascular thrombectomy
IQR	interquartile range
OPTIMAL-BP	Outcome in Patients Treated With Intra-Arterial Thrombectomy-Optimal Blood Pressure Control
SBP	systolic blood pressure

Endovascular thrombectomy (EVT) is an effective therapy for acute ischemic stroke due to large-vessel occlusion, with meta-analyses showing that it significantly increases the likelihood of functional independence compared with medical therapy alone. Persistently elevated blood pressure (BP) after EVT has been associated with poorer outcomes, partly due to higher risks of intracerebral hemorrhage and cerebral edema.¹ Conversely, when the ischemic brain becomes vulnerable to BP fluctuations, BP reduction may lower cerebral perfusion pressure and potentially aggravate ischemic injury.²

To improve functional recovery after successful reperfusion with EVT, 6 recent randomized clinical trials have evaluated the effects of intensive BP lowering during the first 24 to 72 hours after recanalization, using different systolic BP (SBP) targets of <120, 130, 140, or 160 mmHg.^{3–8} Meta-analyses of these trials demonstrated that intensive BP lowering results in worse functional outcomes compared with conventional BP management targeting SBP of 180 mmHg or lower.⁹ Based on these findings, recent guidelines for hypertension and early management of acute ischemic stroke introduced a new recommendation indicating that intensive BP lowering to below 140 mmHg after successful recanalization is harmful.^{10–12}

The association between intensive BP lowering after successful EVT and poor functional outcomes has primarily been attributed to hypoperfusion of penumbral tissue in the setting of impaired cerebral autoregulation.^{4,5,13} However, mechanisms other than cerebral hypoperfusion may also contribute to unfavorable outcomes. EVT necessarily involves the use of iodinated contrast media, which increases the risk of acute kidney injury (AKI).¹⁴ A meta-analysis has reported that AKI occurs in ≈7%

of patients after EVT, and AKI itself is associated with a higher risk of poor functional outcome.¹⁵ Given that hypotension is a recognized risk factor for AKI,¹⁶ intensive BP lowering after EVT, in addition to contrast exposure, may further increase the risk of AKI. Despite this potential link, the effect of postthrombectomy BP management on the development of AKI remains largely unexplored.

In this secondary analysis of the OPTIMAL-BP trial (Outcome in Patients Treated With Intra-Arterial Thrombectomy-Optimal Blood Pressure Control), we aimed to compare the incidence of AKI after successful EVT between intensive and conventional BP management strategies and examine whether the occurrence of AKI was associated with functional dependency or death.

METHODS



Data Availability

The anonymized OPTIMAL-BP data set is available on reasonable request to the corresponding author.

Study Design and Participants

This study is a secondary analysis of the OPTIMAL-BP trial, which was a multicenter, randomized, open-label trial with blinded end-point evaluation conducted at 19 stroke centers in South Korea from June 2020 to November 2022.⁵ The details of the protocol, statistical analysis plan, and patient selection criteria have been reported.^{5,17} Briefly, the trial enrolled patients with acute ischemic stroke due to large-vessel occlusion who attained successful reperfusion of the occluded artery and exhibited an elevated SBP (≥ 140 mmHg) within 2 hours after successful reperfusion.¹⁸ Eligible patients were randomly assigned to receive intensive BP management (SBP target <140 mmHg) or conventional BP management (SBP target 140–180 mmHg) for 24 hours at a 1:1 ratio using a central, password-protected web-based system, stratified by hospital and level of neurological impairment (National Institutes of Health Stroke Scale score at admission, <15 points versus ≥ 15 points). For this analysis, patients with kidney failure requiring replacement therapy before the onset of ischemic stroke and those with missing serum creatinine values were excluded. The Strengthening the Reporting of Observational Studies in Epidemiology guidelines were used to report the findings of this study.¹⁹

Standard Protocol Approvals, Registrations, and Participant Consent

The OPTIMAL-BP trial was registered at ClinicalTrials.gov (NCT04205305). The trial was approved by the institutional

review boards at Severance Hospital (approval number: 4-2019-1208) and all participating centers. Written informed consent was obtained from all participants or their legally authorized representatives before enrollment.

Kidney Function Measures and BP Parameters

Baseline serum creatinine was defined as the most recent value measured within 24 hours before EVT. Serum creatinine levels after EVT were collected at 24-hour intervals from day 1 through a maximum of day 7 after EVT, and missing values were not imputed. Baseline estimated glomerular filtration rate was calculated using the Chronic Kidney Disease Epidemiology Collaboration equation (2009). Proteinuria was evaluated using a urine dipstick test before the EVT. To assess BP-related exposures, BP drop was defined as the occurrence of SBP below 100 mmHg at least once during the first 24 hours after study enrollment. We further defined SBP differential parameters as exposure variables, including the absolute difference and proportional change between the lowest SBP and the SBP at enrollment, as well as between the lowest SBP and the mean SBP during the first 24 hours after enrollment. Additional BP exposure variables included the occurrence of SBP below thresholds of 90, 100, and 110 mmHg, the total duration of time spent below each threshold, and the longest continuous duration below each threshold (Figure S1).²⁰

Outcome Measures

The primary outcome was the occurrence of AKI within 7 days after EVT, defined according to the Kidney Disease: Improving Global Outcomes criteria using the highest available serum creatinine value within each prespecified observation window relative to baseline: an increase in serum creatinine of at least 0.3 mg/dL from baseline, an increase to at least 1.5× the baseline serum creatinine value, or the initiation of kidney replacement therapy. The secondary outcome was the occurrence of AKI within 2 days after EVT, defined using the same criteria. The severity of AKI was further staged based on the maximum increase in serum creatinine relative to baseline, with stage 1 defined as an increase of 1.5 to 1.9× baseline, stage 2 as an increase of 2.0 to 2.9× baseline, and stage 3 as an increase of at least 3.0× baseline or the initiation of kidney replacement therapy.

Post Hoc Analyses According to AKI Status

To evaluate the associations between the occurrence of AKI within 7 days after EVT and functional outcomes, we analyzed prespecified outcomes from the OPTIMAL-BP trial. These included functional independence at 3 months, defined as a modified Rankin Scale score of 0 to 2; stroke-related mortality within 3 months; shift analysis of modified Rankin Scale scores at 3 months; and health-related quality of life assessed using the EuroQoL instrument. The modified Rankin Scale scores were assessed at 3 months through telephone or in-person structured interviews using the Korean version of the modified Rankin Scale (<https://k-stroke-academy.co.kr>), conducted by locally certified medical staff who were blinded to treatment allocation. Health-related quality of life was assessed at 3 months using the 3-Level EuroQoL 5-Dimension Self-Report Questionnaire.²¹

Statistical Analysis

Continuous variables were reported as means with SD or as medians with interquartile ranges (IQRs), and categorical variables were presented as counts and percentages. Comparisons of baseline characteristics between groups were conducted using the Wilcoxon rank-sum test, the χ^2 test, or Fisher exact test, as appropriate. Study outcomes were analyzed using binomial logistic regression models adjusted for the OPTIMAL-BP trial covariates (age, sex, time from stroke onset to enrollment, and the National Institutes of Health Stroke Scale score at admission), with an additional model further adjusting for baseline estimated glomerular filtration rate. Adjusted odds ratios (AORs) with 95% CIs were reported. As an additional sensitivity analysis, we repeated the 48-hour AKI analysis under equal ascertainment conditions by restricting the analysis to patients with serum creatinine measurements available at both prespecified post-EVT time points, day 1 and day 2. AKI was then redefined according to the Kidney Disease: Improving Global Outcomes creatinine criteria using the higher of the day 1 and day 2 creatinine values relative to baseline.

Post hoc analyses were performed to evaluate the association between the occurrence of AKI within 7 days after EVT and clinical outcomes. Associations between AKI and clinical outcomes were evaluated using logistic regression models. AORs with 95% CIs were estimated using the same covariates used in the analysis of the primary and secondary outcomes. To assess whether AKI mediated the association between intensive BP management and functional dependency or mortality at 3 months, a 4-way causal mediation analysis was performed using the med4way package in STATA 18.0 (StataCorp, College Station, TX). A 4-way decomposition model was used to partition the total effect into 4 components: the controlled direct effect (neither mediation nor interaction), reference interaction (interaction only), mediated interaction (both mediation and interaction), and pure indirect effect (mediation only). Point estimates, standard errors, and 95% CIs for each of the 4 components were obtained using the delta method, as implemented in the med4way package. To quantify the contribution of the mediator to the overall association, the proportion of the pure indirect effect relative to the total effect was calculated. To assess the robustness of key associations to potential unmeasured confounding, we calculated *E*-values using the *E*-value package in R. The *E*-value indicates how strong an unmeasured confounder would need to be, in its associations with both the exposure and the outcome, to completely account for the observed association. *E*-values were calculated for the point estimates and for the CI limits closest to the null.

To identify clinical variables associated with the occurrence of AKI within 7 days after EVT, univariable analyses were performed between AKI and clinical variables, including treatment assignment (intensive or conventional BP management). Variables with a *P* < 0.1 in univariable analyses were entered into a multivariable adjusted model to identify factors independently associated with AKI. Finally, to examine the association between BP reduction-related variables and AKI, multivariable logistic regression analyses were conducted for each BP parameter, adjusting for the variables that remained statistically significant in the preceding multivariable model.

Because the BP-related exposure variables were highly collinear, they were not included simultaneously in the same

multivariable model. Instead, each BP parameter was entered individually into a separate multivariable logistic regression model using the same non-BP covariate adjustment set. We did not apply a multiplicity correction given the nature of the post hoc analysis. Analyses were performed using R, version 4.5.2 (R Foundation for Statistical Computing, Vienna, Austria), and STATA 18.0 (StataCorp, College Station, TX).

RESULTS

Study Population and Baseline Characteristics

Among the 306 enrolled patients, 19 patients were excluded; 1 withdrew consent after randomization, 1 had kidney failure requiring replacement therapy, 2 lacked baseline serum creatinine data, and 15 lacked follow-up serum creatinine measurements during hospitalization (Figure 1). We compared patients with and without follow-up creatinine data to assess potential informative missingness. Baseline characteristics were generally similar, except for a longer time from stroke onset to enrollment in patients without follow-up creatinine data (Table S1). The final cohort comprised 287

patients (mean age, 73.2 [SD, 11.6] years; 117 [40.8%] women), of whom 147 were assigned to the intensive BP management group and 140 to the conventional BP management group. Baseline characteristics were well balanced between the 2 groups. The mean National Institutes of Health Stroke Scale score at admission was 13.0 (SD, 6.1; Table 1). Kidney function-related variables did not significantly differ between groups, including baseline estimated glomerular filtration rate (intensive group, 81.5 [IQR, 66.3–92.3] mL/min per 1.73 m² versus conventional group, 79.9 [IQR, 65.0–92.7] mL/min per 1.73 m²; $P=0.501$) and the presence of proteinuria. During the 7-day hospitalization period, serum creatinine was measured more frequently in the intensive management group than in the conventional management group (median, 5.0 [IQR, 3.0–7.0] versus 4.0 [IQR, 3.0–6.0]; $P=0.014$). There were no significant between-group differences in the use of renin-angiotensin system blockers, diuretics, or statins before admission, and the distribution of detailed pre-EVT imaging modalities was also similar between the 2 groups (Table 2). All patients underwent EVT under conscious sedation.

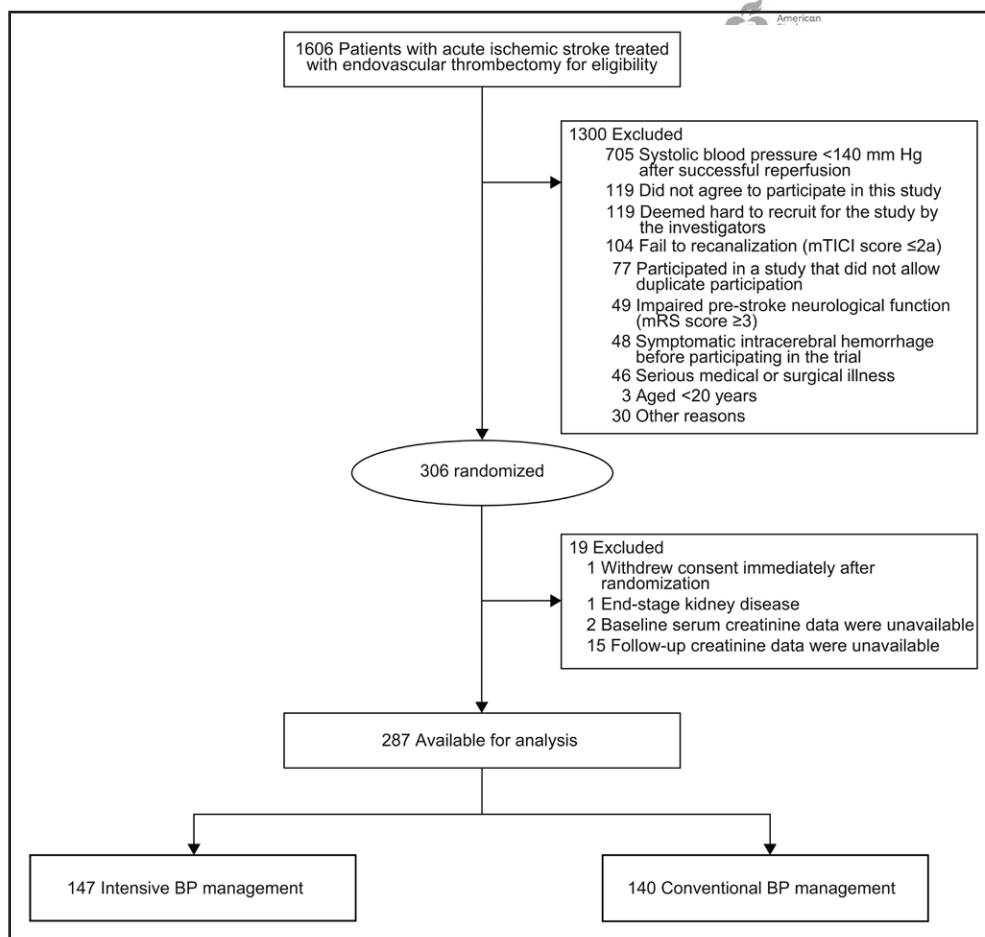


Figure 1. Patient flowchart.

Of 1606 patients with acute ischemic stroke treated with endovascular thrombectomy, 306 were randomized. After exclusion of 19 patients, 287 were included in the final analysis: 147 in the intensive blood pressure (BP) management group and 140 in the conventional BP management group. mRS indicates modified Rankin Scale; and mTICI, modified Thrombolysis in Cerebral Infarction.

Table 1. Baseline Characteristics

Values*	Conventional management (n=140)	Intensive management (n=147)	P value†
Demographics			
Age, y; mean (SD)	73.2 (10.8)	73.2 (12.3)	0.557
Sex			0.796
Male	84 (60.0)	86 (58.5)	
Female	56 (40.0)	61 (41.5)	
Medical history			
Hypertension	104 (74.3)	114 (77.6)	0.518
Diabetes	60 (42.9)	63 (42.9)	>0.999
Hyperlipidemia	52 (37.1)	55 (37.4)	0.962
Coronary artery obstructive disease	16 (11.4)	17 (11.6)	0.971
Atrial fibrillation	64 (46.0)	76 (51.7)	0.339
Congestive heart disease	7 (5.0)	7 (4.8)	0.925
Previous stroke history	30 (21.4)	34 (23.1)	0.729
Active cancer	5 (3.6)	9 (6.1)	0.316
Smoking	27 (19.3)	37 (25.3)	0.231
IV-tPA use	51 (36.4)	42 (28.6)	0.155
NIHSS score at admission, mean (SD)	12.7 (6.4)	13.3 (5.8)	0.227
Radiological and procedural variables			
Occlusion site			0.604
MCA	96 (68.6)	97 (66.0)	
ICA	28 (20.0)	29 (19.7)	
VBA	10 (7.1)	16 (10.9)	
ACA	3 (2.1)	1 (0.7)	
ICA+MCA	2 (1.4)	4 (2.7)	
PCA	1 (0.7)	0 (0.0)	
ASPECTS, median (IQR; n=283)	9 (8–10)	9 (7–10)	0.277
mTICI score (immediate reperfusion)			0.672
2b	32 (22.9)	33 (22.4)	
2c	22 (15.7)	18 (12.2)	
3	86 (61.4)	96 (65.3)	
Time parameters, median (IQR), min			
Onset to puncture time	356.0 (208.0–711.0)	386.0 (222.0–680.0)	0.702
Onset to reperfusion time	428.0 (271.5–768.5)	430.0 (270.0–739.0)	0.878
Onset to enrollment time	472.5 (312.5–815.0)	494.0 (310.0–795.0)	0.901

ACA indicates anterior cerebral artery; ASPECTS, Alberta Stroke Program Early Computed Tomography Score; ICA, internal carotid artery; IQR, interquartile range; IV, intravenous; MCA, middle cerebral artery; mTICI, modified Treatment in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; PCA, posterior cerebral artery; SBP, systolic blood pressure; tPA, tissue-type plasminogen activator; and VBA, vertebrobasilar artery.

*Values are medians (IQR), mean (SD) or number (%).

†P value <0.05 was considered of statistical significance.

BP Profiles

The mean SBP at enrollment was 155.1 (SD, 14.7) mmHg in the conventional management group and 154.8 (SD, 13.4) mmHg in the intensive management group. The intensive management group had significantly lower 24-hour mean SBP (129.1 [SD, 7.8] mmHg versus 137.5 [SD, 13.7] mmHg; $P<0.001$) and diastolic BP (72.1 [SD, 7.9] mmHg versus 76.6 [SD, 9.7] mmHg;

$P<0.001$) than the conventional management group (Figure 2). Within the first 24 hours after randomization, intravenous antihypertensive therapy was administered more frequently in the intensive management group than in the conventional management group (72.1% versus 19.3%; $P<0.001$), which also experienced a higher frequency of SBP drops below 100 mmHg (38.8% versus 19.3%; $P<0.001$; Table 2).

Table 2. Baseline Characteristics Regarding Kidney Function

Values*	Conventional management (n=140)	Intensive management (n=147)	P value†
Kidney function-related variables			
eGFR, mL/min per 1.73 m ² ; median (IQR)	79.9 (65.0–92.7)	81.5 (66.3–92.3)	0.501
eGFR stratification, mL/min per 1.73 m ²			0.910
>90	42 (30.0)	45 (30.6)	
60–89	68 (48.6)	72 (49.0)	
30–59	27 (19.3)	25 (17.0)	
<30	3 (2.2)	5 (3.4)	
Urine dipstick for proteinuria (n=275)			0.825
Negative	90/134 (67.2)	88/141 (62.4)	
Trace	20/134 (14.9)	22/141 (15.6)	
1+	15/134 (11.2)	20/141 (14.2)	
2+	7/134 (5.2)	10/141 (7.1)	
3+	2/134 (1.5)	1/141 (0.7)	
No. of Cr measurement, median (IQR)	4.0 (3.0–6.0)	5.0 (3.0–7.0)	0.014
Medications before admission			
RAS blockade	35 (25.0)	43 (29.3)	0.418
Diuretics	21 (15.0)	15 (10.2)	0.220
Statin	38 (27.1)	43 (29.3)	0.692
Pre-EVT imaging modality (use of iodinated contrast)			0.426
CT angiography with CT perfusion	104 (74.3)	108 (73.5)	
CT angiography	22 (15.7)	18 (12.2)	
No iodinated contrast use (MR or noncontrast CT)	14 (10.0)	21 (14.3)	
BP and procedure related variables			
SBP at enrollment, mean (SD), mm Hg	155.1 (14.7)	154.8 (13.4)	0.748
DBP at enrollment	85.4 (14.2)	83.9 (14.5)	0.254
SBP at 1 h	140.0 (19.8)	135.2 (20.4)	0.038
DBP at 1 h	79.6 (14.7)	78.2 (16.2)	0.332
SBP at 24 h	136.8 (20.0)	128.3 (16.5)	<0.001
DBP at 24 h	75.0 (11.7)	70.8 (12.7)	0.003
Mean SBP over 24 h	137.5 (13.7)	129.1 (7.8)	<0.001
Mean DBP over 24 h	76.6 (9.7)	72.1 (7.9)	<0.001
IV antihypertensives during study period	27 (19.3)	106 (72.1)	<0.001
Nicardipine	27 (19.3)	98 (66.7)	<0.001
Labetalol	0 (0.0)	10 (6.8)	0.002
Both	0 (0.0)	2 (1.4)	
BP drop (<SBP 100 mm Hg)	27 (19.3)	57 (38.8)	<0.001
Device passing number, median (IQR)	1.0 (1.0–2.0)	1.0 (1.0–2.0)	0.591
Puncture to reperfusion time, min; median (IQR)	45.5 (32.5–80.0)	44.0 (29.0–83.0)	0.566

BP indicates blood pressure; Cr, creatinine; CT, computed tomography; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; EVT, endovascular thrombectomy; IQR, interquartile range; IV, intravenous; MR, magnetic resonance; RAS, renin-angiotensin system; and SBP, systolic blood pressure.

*Values are medians (IQR), mean (SD) or number (%).

†P value <0.05 was considered of statistical significance.

Kidney Outcomes

The incidence of AKI within 7 days was significantly higher in the intensive management group than in the conventional management group (intensive group, 13.6% versus conventional group, 6.4%; AOR, 2.54 [95% CI,

1.10–6.35]; $P=0.036$). The occurrence of stage 1 AKI was more frequent in the intensive management group, whereas the incidences of stage 2 and stage 3 AKI did not significantly differ between the 2 groups. Similarly, AKI within 2 days occurred more frequently in the intensive

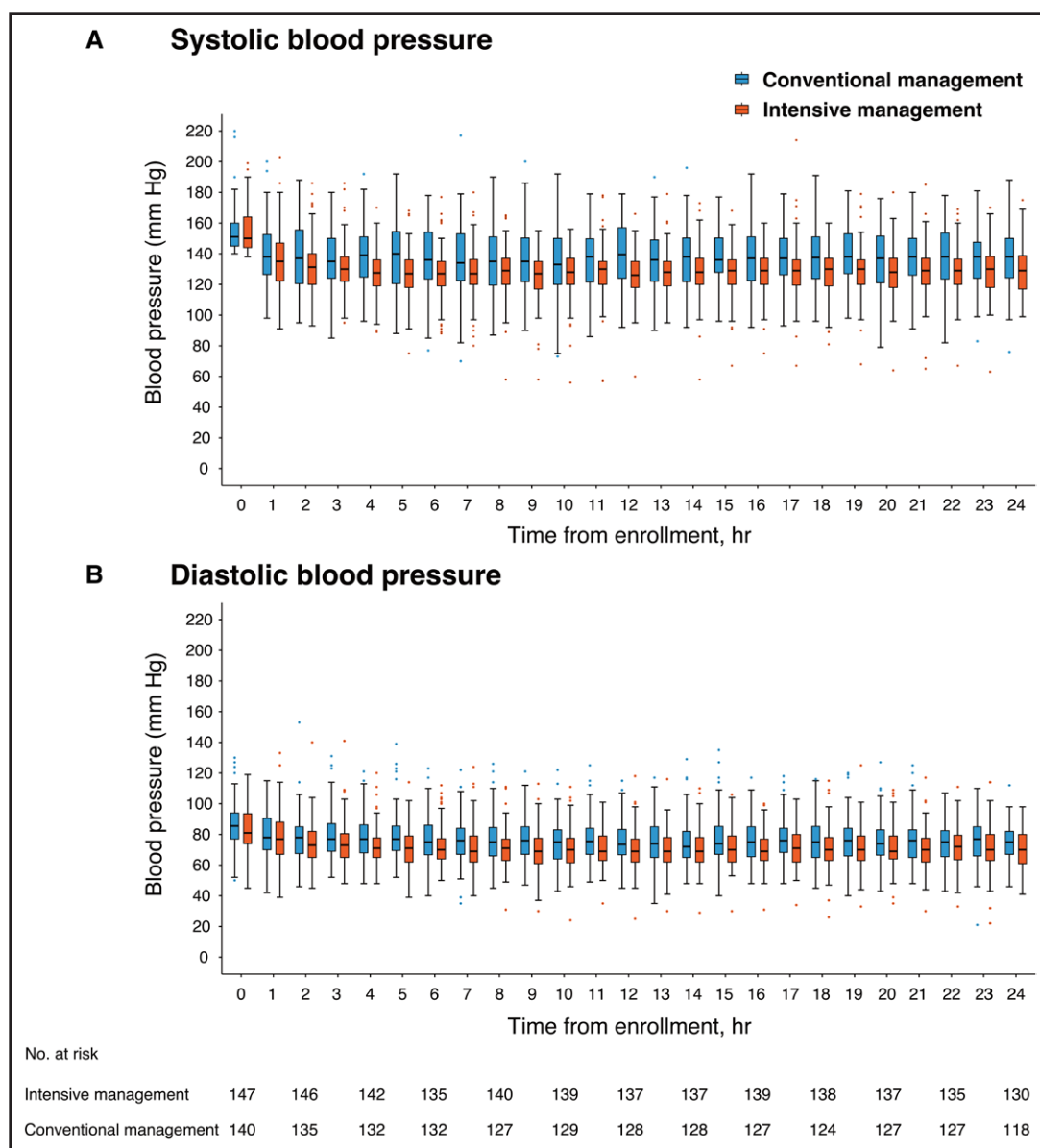


Figure 2. Changes in mean systolic and diastolic blood pressure (BP) from randomization to 24 hours.

Box-and-whisker plots depict hourly systolic (A) and diastolic (B) BP measurements in the intensive and conventional BP management groups. The box represents the interquartile range (25th to 75th percentiles), with the median indicated by the horizontal line. Numbers at risk at each time point are shown.

management group than in the conventional management group (intensive group, 6.9% versus conventional group, 1.5%; AOR, 7.20 [95% CI, 1.58–55.40]; $P=0.023$). As observed for the 7-day outcome, stage 1 AKI was more common in the intensive management group, while no significant between-group differences were found in the incidences of stage 2 and stage 3 AKI (Table 3).

Sensitivity Analysis of 48-Hour AKI Under Equal Creatinine Ascertainment

To address potential differential AKI ascertainment due to more frequent creatinine testing in the intensive BP management group, we performed an additional

sensitivity analysis restricted to patients with serum creatinine measurements available at both post-EVT day 1 and day 2 (Table S2). Among the 142 patients meeting this criterion, baseline characteristics were generally balanced between the conventional and intensive management groups, although diabetes tended to be more frequent in the conventional group (Table S3). In this equal-ascertainment restricted cohort, AKI within 48 hours occurred in 2 of 66 (3.0%) patients in the conventional management group and 7 of 76 patients (9.2%) in the intensive management group. The adjusted odds ratio was 4.20 (95% CI, 0.83–32.64; $P=0.108$; Table S4). Although statistical significance was not reached because of the reduced sample size, the absolute risk

Table 3. Kidney Outcomes

	Conventional management (n=140)	Intensive Management (n=147)	Unadjusted effect size (95% CI)	Adjusted effect size (95% CI; model 1)*	Adjusted effect size (95% CI; model 2)†	P value‡
AKI (within 48 h)	2/135 (1.5)	10/144 (6.9)	4.96 (1.28–32.69)	5.57 (1.35–38.32)	7.20 (1.58–55.40)	0.023
Stage 1	1/135 (0.7)	8/144 (5.6)	7.88 (1.42–147.23)	9.16 (1.53–177.10)	10.2 (1.64–202.85)	0.038
Stage 2	1/135 (0.7)	0/144 (0.0)	...	...	...	...
Stage 3	0/135 (0.0)	2/144 (1.4)	...	...	...	...
AKI (within 7 d)	9/140 (6.4)	20/147 (13.6)	2.29 (1.03–5.47)	2.30 (1.02–5.58)	2.54 (1.10–6.35)	0.036
Stage 1	6/140 (4.3)	14/147 (9.5)	2.35 (0.91–6.81)	2.63 (0.98–7.95)	2.91 (1.06–9.06)	0.048
Stage 2	0/140 (0.0)	3/147 (2.0)	...	...	...	...
Stage 3	3/140 (2.1)	3/147 (2.0)	0.95 (0.17–5.22)	0.94 (0.17–5.28)	1.01 (0.17–6.34)	0.987

AKI indicates acute kidney injury; eGFR, estimated Glomerular Filtration Rate; and NIHSS, National Institutes of Health Stroke Scale.

*Model 1 was adjusted for age, sex, time from stroke onset to enrollment, and NIHSS score at admission.

†Model 2 was additionally adjusted for baseline eGFR on top of the covariates included in Model 1.

‡The reported P values correspond to the adjusted effect estimates from Model 2.

difference was 6.2%, which was slightly larger than that observed in the main 48-hour analysis, indicating a directionally consistent pattern under matched ascertainment conditions.

Clinical Outcomes According to AKI Status

In the analyses stratified by the occurrence of AKI within 7 days after EVT, patients who developed AKI had a higher prevalence of diabetes and congestive heart disease and a lower baseline estimated glomerular filtration rate compared with those who did not develop AKI (Table S5). Patients who developed AKI within 7 days had a significantly lower rate of functional independence at 3 months (13.8% versus 49.0%; AOR, 0.19 [95% CI, 0.05–0.55]; $P=0.005$) and a significantly higher rate of stroke-related mortality within 3 months (37.9% versus 3.1%; AOR, 13.8 [95% CI, 4.14–49.64]; $P<0.001$) than those without AKI (Figure 3). Health-related quality of life, assessed using the 3-Level EuroQol 5-Dimension Self-Report

Questionnaire, was also significantly lower in patients who developed AKI within 7 days compared with those who did not (Table 4).

In an exploratory 4-way causal mediation analysis, the association between intensive BP management and functional dependency showed a complex pattern. While the pure indirect effect through AKI was statistically significant but modest (pure indirect effect: β , -0.067 [95% CI, -0.126 to -0.008]; $P=0.026$), accounting for 11.6% of the total effect, the reference interaction component was substantially larger (reference interaction: β , -0.753 [95% CI, -1.057 to -0.450]; $P<0.001$). This pattern suggests that the association between intensive BP management and functional outcome may differ according to the presence of AKI, while AKI may also partially mediate this relationship (Table S6). To assess the robustness of the observed associations to unmeasured confounding, E-values were calculated for the main outcomes (Table S7). For the association between intensive BP management and 7-day AKI, the E-value was 4.52 for the point estimate but only 1.43 for the CI limit closest to the null.

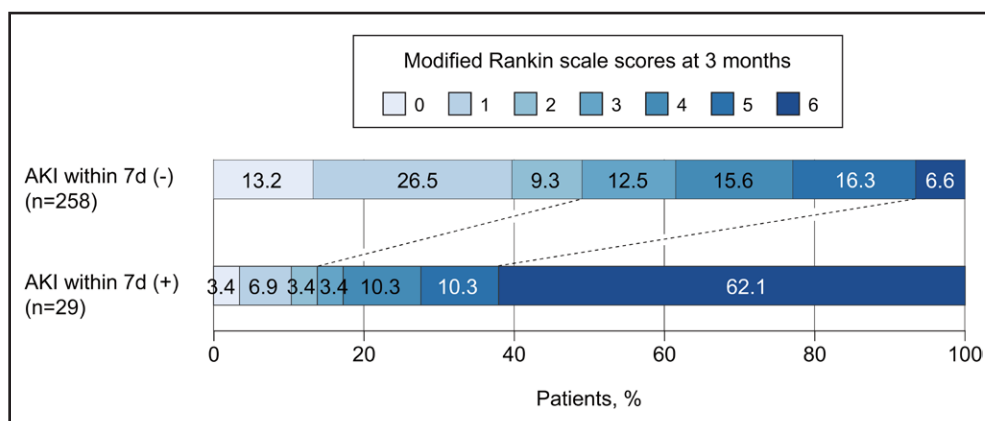


Figure 3. modified Rankin Scale (mRS) distribution at 3 months according to the presence of acute kidney injury (AKI) within 7 days.

Stacked bar charts depict the distribution of mRS scores at 3 months in patients with and without AKI within 7 days. Values are expressed as percentages.

Table 4. Efficacy and Safety Outcomes According to the Presence of AKI

	AKI within 7 d (+; n=29)*	AKI within 7 d (−; n=258)*	Unadjusted effect size (95% CI)	Adjusted effect size (95% CI)	P value
Primary outcomes					
mRS 0–2 (functional independence) at 3 mo†	4/29 (13.8)	126/257 (49.0)	0.17 (0.05–0.44)	0.19 (0.05–0.55)	0.005
Mortality related to the stroke within 3 mo†	11/29 (37.9)	8/257 (3.1)	19.0 (6.89–55.23)	13.8 (4.14–49.64)	<0.001
Secondary outcomes					
Ordinal analysis of category scores on the mRS at 3 mo†	6 (4–6)	3 (1–4)	14.7 (6.50–35.22)	16.8 (6.91–43.60)	<0.001
Euro-QoL score‡	0.00 (0.00–0.38)	0.72 (0.10–0.91)	−0.38 (−0.54 to −0.22)	−0.28 (−0.43 to −0.14)	<0.001

Multivariable analysis of outcomes adjusted for age, sex, NIHSS at admission, onset to enrollment time, and baseline eGFR. AKI indicates acute kidney injury; eGFR, estimated glomerular filtration rate; Euro-QoL, 3-Level Euro-QoL 5-Dimension Self-Report Questionnaire; IQR, interquartile range; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale.

*Data are presented as number (percentage) of patients for categorical variables and median (IQR) for continuous or ordinal variables.

†The effect size is reported for the primary outcomes as an odds ratio with the 95% CI.

‡For mRS score reduction, as a generalized odds ratio with the 95% CI.

§For Euro-QoL score, as beta coefficient with 95% CI.

BP Profiles Associated With AKI

Patients who developed AKI exhibited distinct BP profiles during the first 24 hours after EVT. Although SBP at enrollment and mean SBP during the first 24 hours did not differ between patients with and without AKI, those who developed AKI showed parameters reflecting greater SBP reduction (mean lowest SBP during the first 24 hours: 96.6 [18.4] mmHg versus 106.4 [12.9] mmHg; $P=0.004$). Furthermore, hypotension events were more frequent and prolonged among patients with AKI (Table S8). Episodes of SBP <100 mmHg occurred in 48.3% of patients with AKI versus 27.1% of those without AKI ($P=0.02$), and the cumulative time spent below an SBP of 100 mmHg was nearly 4-fold longer in the AKI group (114.9 ± 240.4 min versus 29.5 ± 85.7 min; $P=0.007$; Table S8). To identify factors associated with the development of AKI, a multivariable analysis incorporating relevant clinical covariates demonstrated that intensive BP management and diabetes were independently associated with the occurrence of AKI within 7 days (Table S9). Lower SBP, greater SBP reductions, and increased frequency and duration of hypotensive episodes across multiple thresholds (<90, <100, and <110 mmHg) remained independently associated with AKI (Table S10).

DISCUSSION

In this secondary analysis of the OPTIMAL-BP trial, intensive BP management targeting SBP <140 mmHg after EVT was associated with a significantly higher incidence of AKI compared with conventional BP management. Despite the mild severity of AKI, its development was strongly associated with worse functional outcomes, higher stroke-related mortality, and poorer health-related quality of life at 3 months. Exploratory mediation analysis suggested that AKI may play a complex role in the relationship between intensive BP management and

functional outcomes, although the precise nature of this relationship remains uncertain. Collectively, these findings suggest that AKI may contribute to the association between intensive post-EVT BP lowering and adverse clinical outcomes, although further prospective studies are needed.

Several randomized trials and meta-analyses have consistently shown that intensive BP lowering after successful EVT does not improve, and may worsen, functional outcomes.^{4,5,9} Although cerebral hypoperfusion due to impaired autoregulation has been proposed as the primary mechanism,¹³ our findings extend this paradigm by highlighting the kidney as an additional end-organ vulnerable to aggressive BP reduction. In the present study, the incidence of AKI was more than 2-fold higher in the intensive BP management group, despite comparable baseline renal function and procedural characteristics between groups. Previous studies have shown that even mild and transient elevations in serum creatinine are associated with unfavorable short- and long-term outcomes, particularly in older patients with vascular comorbidities, although most AKI events are classified as stage 1.^{22,23} Consistently, even predominantly stage 1 AKI in our cohort was associated with reduced functional independence and increased stroke-related mortality, highlighting the clinical relevance of mild AKI after EVT. Accordingly, the observed increase in mainly mild AKI is clinically meaningful and supports a cautious approach to intensive BP reduction after successful EVT.

Beyond treatment assignment, AKI was closely associated with BP profiles reflecting excessive SBP reduction, including lower nadir SBP, greater magnitude of SBP decline, and longer duration of hypotensive exposure during the first 24 hours after EVT. Indeed, mean SBP values alone were not discriminative, emphasizing that dynamic BP fluctuations and transient hypotension may be more critical determinants of kidney injury than average BP levels. From a mechanistic perspective, large-vessel occlusion stroke is associated with

central autonomic dysfunction, leading to heightened sympathetic activation and concomitant parasympathetic withdrawal.²⁴ This neurohumoral state may impair renal autoregulatory capacity,²⁵ rendering the kidneys more vulnerable to abrupt BP reductions induced by intravenous antihypertensive agents. These findings suggest that repeated or sustained excursions below renal autoregulatory thresholds could precipitate AKI.²⁶ Such hemodynamic vulnerability may not be fully captured by current BP targets, as transient BP reductions are often difficult to detect with conventional intermittent BP monitoring. Accordingly, BP management after EVT should place greater emphasis on hypotensive burden, and more granular BP monitoring strategies may improve detection of clinically meaningful hypotensive events.

Patients who developed AKI within 7 days after EVT had markedly lower rates of functional independence and substantially higher stroke-related mortality at 3 months. These findings are consistent with prior observational studies linking AKI to poor outcomes after acute ischemic stroke.^{15,27,28} The occurrence of AKI can, in part, reflect greater overall illness severity and reduced physiological reserve, thereby identifying patients who are intrinsically vulnerable to adverse neurological recovery.²⁹ However, AKI itself may also actively contribute to poor outcomes through kidney-brain interactions, including the amplification of systemic inflammation, endothelial dysfunction, oxidative stress, and disturbances in volume and electrolyte homeostasis, all of which adversely affect cerebral recovery potential.^{30,31} In line with these 2 possibilities, our 4-way decomposition showed that the association between intensive BP management and poor functional outcomes was explained predominantly by a treatment-AKI interaction. That is, the occurrence of AKI identified a vulnerable subgroup of patients in whom intensive BP lowering was particularly harmful. Any mediation through AKI appeared to be relatively minor, suggesting that kidney-brain interactions contributed only modestly to the overall association.

Contrast exposure remains an important competing explanation for AKI after EVT.³² Although prior studies have suggested that high contrast volume and repeated iodinated contrast exposure may increase AKI risk,³³ evidence in the EVT setting has been inconsistent.^{22,34} Against this background, the similar pre-EVT imaging strategies, EVT procedure time, and device pass number between groups in the present study make it unlikely that differences in contrast exposure alone fully account for the higher AKI incidence observed in the intensive BP management group. These findings instead suggest that post-EVT hemodynamic factors, particularly greater SBP reduction and more frequent hypotensive exposure, may have additionally contributed to kidney injury in this setting.

An important clinical question is whether AKI in this setting represents a modifiable therapeutic target or primarily a marker of hemodynamic instability. Our findings

suggest that both interpretations may be relevant. AKI may identify patients with reduced physiological reserve who are particularly vulnerable to excessive BP reduction after EVT, while, as previously discussed, kidney injury itself may also contribute to adverse outcomes. Therefore, AKI should not be viewed solely as a direct therapeutic target based on the present analysis, but rather as a clinically meaningful indicator of systemic vulnerability and a possible contributor to poor recovery. Whether prevention of AKI itself can improve neurological outcomes requires confirmation in future studies.

Our findings have important clinical implications. First, these findings suggest that aggressive BP lowering after EVT may cause systemic harm beyond the brain, particularly to the kidneys, underscoring the need to reconsider postreperfusion BP management as a multiorgan issue rather than a brain-exclusive strategy. In the acute post-EVT period, patients are exposed to multiple renal stressors, such as contrast media, hemodynamic instability, and neurohumoral activation, which may collectively increase susceptibility to excessive BP reduction. Second, even mild AKI should not be dismissed as clinically insignificant. Although most AKI events observed in this study were classified as stage 1, their occurrence was strongly associated with functional dependency and mortality at 3 months. Finally, BP management strategies after EVT should aim to preserve adequate perfusion to both the brain and kidneys, recognizing that excessive BP reductions may be detrimental to these vulnerable organs. Rather than relying solely on fixed SBP targets, future approaches may benefit from incorporating dynamic BP parameters, such as hypotensive burden, BP variability, and nadir BP levels, to better balance the competing risks of cerebral hypoperfusion and systemic organ injury. Taken together, these considerations underscore the importance of individualized BP management after EVT, informed by both neurological and systemic perfusion requirements.

This study has several limitations. First, this analysis was a secondary analysis of a randomized clinical trial, and therefore causal inferences should be interpreted with caution. In particular, the 4-way causal mediation analysis requires strong assumptions including no unmeasured confounding of exposure-mediator, exposure-outcome, and mediator-outcome relationships, as well as no mediator-outcome confounders affected by the exposure. These assumptions are difficult to verify in observational data, and violations could substantially bias the estimates of mediation effects. This limitation is further reflected by the E-value for the CI limit closest to the null, which was only 1.43. In a setting where residual imbalance in baseline AKI risk, contrast exposure, and illness severity cannot be excluded, such a relatively modest E-value suggests that the primary association between intensive BP management and AKI is of limited robustness to potential unmeasured confounding. Therefore, our findings should be considered exploratory rather than definitive evidence

of causal mechanisms. Second, AKI was defined solely on the basis of serum creatinine criteria, as urine output data were not uniformly available across study sites, which may have led to an underestimation of AKI incidence. Moreover, the equal-ascertainment sensitivity analysis was feasible only for the 48-hour end point. A comparable analysis for the 7-day primary end point could not be reliably performed because follow-up creatinine measurements became increasingly incomplete and imbalanced between the treatment groups after day 2. Thus, some uncertainty regarding differential ascertainment of the 7-day AKI outcome remains. Third, AKI after EVT may be related in part to contrast exposure, but detailed quantitative data on total contrast volume were not available in the present study. The distribution of pre-EVT imaging modalities was similar between the intensive and conventional BP management groups, suggesting that a major imbalance in preprocedural contrast exposure is unlikely to fully explain the observed between-group difference in AKI incidence. Nonetheless, because exact total contrast load was unavailable, residual confounding related to contrast burden cannot be excluded. Fourth, the relatively small number of AKI events may have resulted in instability of statistical models, as reflected by wide 95% CIs, particularly in subgroup and mediation analyses. Nevertheless, the direction and magnitude of effect estimates were largely consistent across analytic approaches, supporting the robustness of the observed associations. Fifth, consistent with earlier studies, only a small proportion of patients had severe baseline renal impairment, which may restrict the applicability of our findings to patients with advanced chronic kidney disease. However, this also reflects real-world EVT populations commonly enrolled in contemporary randomized trials. Sixth, intraprocedural BP data were not collected in the OPTIMAL-BP data set. Therefore, we could not evaluate the potential contribution of intraprocedural hemodynamic fluctuations to the development of early AKI. Seventh, the association between AKI and stroke-related mortality should be interpreted cautiously. The markedly higher mortality observed in patients with AKI may partly reflect greater overall illness severity and cooccurring treatment limitation or goals-of-care decisions, rather than a purely direct effect of AKI itself. Because these data were not systematically collected, they could not be adjusted for in the outcome models, and residual confounding cannot be excluded. Finally, because the OPTIMAL-BP trial was conducted exclusively in a Korean population, the generalizability of these findings to other ethnic and geographic populations may be limited.

CONCLUSIONS

In patients undergoing successful EVT, intensive BP lowering was associated with a higher risk of AKI, predominantly mild in severity but clinically consequential. AKI

after EVT was strongly associated with worse functional outcomes, higher mortality, and poorer quality of life. These findings highlight AKI as an important systemic complication of aggressive postthrombectomy BP lowering and suggest that kidney protection could be considered alongside cerebral perfusion when defining optimal BP targets after EVT.

ARTICLE INFORMATION

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Supplemental Material

Checklist
Tables S1–S10
Figure S1

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